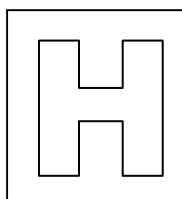


Candidate Name: _____

Class Adm No

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2017 Preliminary Examination 2

Pre-University 3

H2 Biology

9648/02

Paper 2 Core Paper

13 September 2017

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

The use of an approved scientific calculator is expected, where appropriate. You will lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

This question paper consists of 23 printed pages.

[Turn over

Section A

Answer **all** questions in this section.

1. Fig. 1.1 shows a series of micrographs of animal cells undergoing cell and nuclear division.

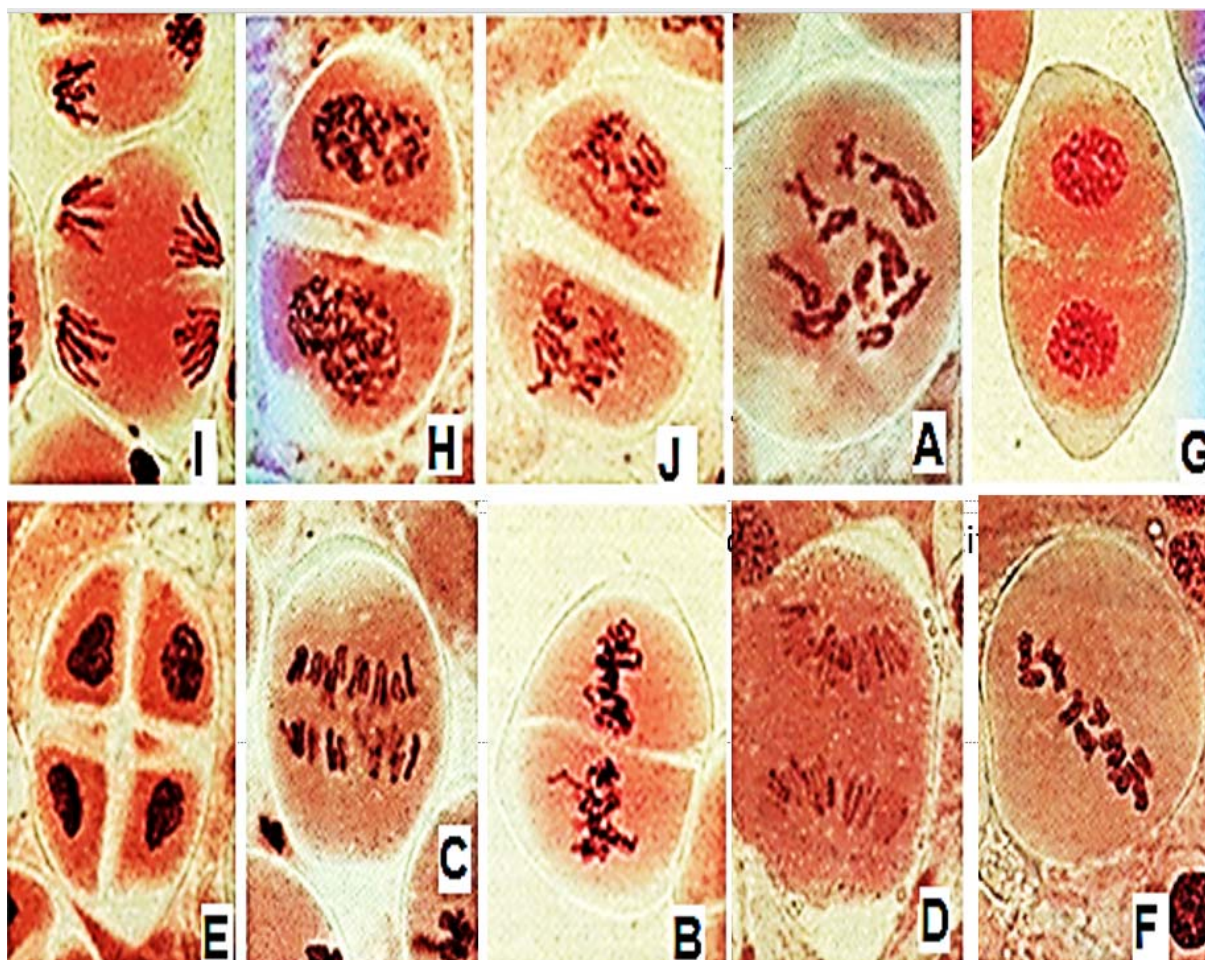


Fig. 1.1

- (a) Arrange the letters in Fig. 1.1 in a correct sequence to show the events occurring in the cell and nuclear division process.

.....[2]

- (b) Describe the processes occurring in **A**.

.....

[2]

- (c) State one similarity and one difference between process **C** and **J**.

.....

.....

.....

.....[2]

- (d) Explain the significance of process **H**.

.....

.....

.....

.....[2]

- (e) Suggest how the cell and nuclear division process would be affected if centromeric DNA is deleted from a chromosome.

.....

.....

.....

.....[2]

[Total 10]

2. In bacteria, the production of the amino acid tryptophan is catalyzed by five specific enzymes (simply named as **E**, **D**, **C**, **B** and **A** in this question) encoded by specific genes *trpE*, *trpD*, *trpC*, *trpB* and *trpA*. The *trp* operon is transcriptionally regulated by a repressor protein, (named **R** in this question), encoded by the *trpR* gene. Expression of the *trpE*, *trpD*, *trpC*, *trpB* and *trpA* genes is controlled by a promoter region and an operator region.

When levels of tryptophan are high, tryptophan binds to the repressor protein, **R**. The tryptophan-repressor protein complex binds to the operator region and prevents expression of the *trpE*, *trpD*, *trpC*, *trpB* and *trpA* genes.

- (a) Draw a simple diagram to show the *trp* operon.

[1]

- (b) Explain why it is useful for a bacterial cell to decrease expression of the *trp* genes when tryptophan is present.

.....
.....
.....
.....[2]

Table 2.1 below indicates the activity levels of the functional enzymes E, D, C, B and A in wild type bacterial cells in the presence and absence of tryptophan (Trp).

Table. 2.1

Enzyme	Activity level of enzymes/units	
	Trp absent	Trp present
E	700	0
D	700	0
C	700	0
B	700	0
A	700	0

Researchers have managed to obtain several bacterial mutants. Each mutant is the result of a single base-pair substitution in a single component of the *trp* operon. The activity level of functional enzymes E, D, C, B and A in the bacterial cells having these individual mutations is shown in Table 2.2.

Table. 2.2

Enzymes	Activity level of enzymes/units					
	Mutant 1		Mutant 2		Mutant 3	
	Trp absent	Trp present	Trp absent	Trp present	Trp absent	Trp present
E	700	700	700	0	0	0
D	700	700	0	0	0	0
C	700	700	700	0	0	0
B	700	700	700	0	0	0
A	700	700	700	0	0	0

- (c) With reference to Table 2.1 and Table 2.2, identify the mutant bacteria that has a phenotype that is consistent with a loss-of-function mutation in the *trpR* gene and explain your choice.

.....

.....

.....

.....[2]

- (d) Assuming mutant 3 experienced a loss-of-function mutation, account for its phenotype.

.....
.....
.....
.....[2]

- (e) If the phenotype of mutant 3 is caused by a mutation in the *trpR* gene, explain how this mutation would affect the structure and function of the repressor protein.

.....
.....
.....
.....[2]

[Total: 9]

3. Researchers are constantly investigating the effects of limiting factors on the rate of photosynthesis on various plants. Fig. 3.1 show how three main limiting factors, carbon dioxide concentration, light intensity and temperature can affect the rate of photosynthesis in cactus plants.

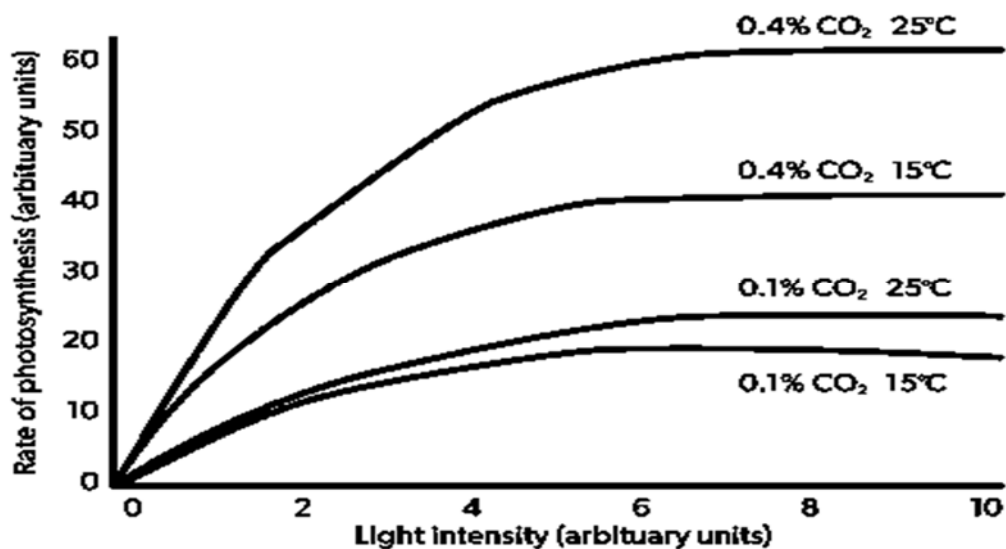


Fig. 3.1

- (a) Define the term 'limiting factor'.

.....
[1]

- (b) With reference to Fig. 3.1,

- (i) explain for the effect of light intensity on rate of photosynthesis.

.....

[5]

- (ii) justify if carbon dioxide concentration or temperature is a greater limiting factor on the rate of photosynthesis.

.....

[3]

- (c) Suggest why water is not considered a limiting factor for the rate of photosynthesis.

.....
[1]

Fig. 3.2 illustrates a graph showing how varying light intensity affects the net carbon dioxide uptake and release in sun and shade plants.

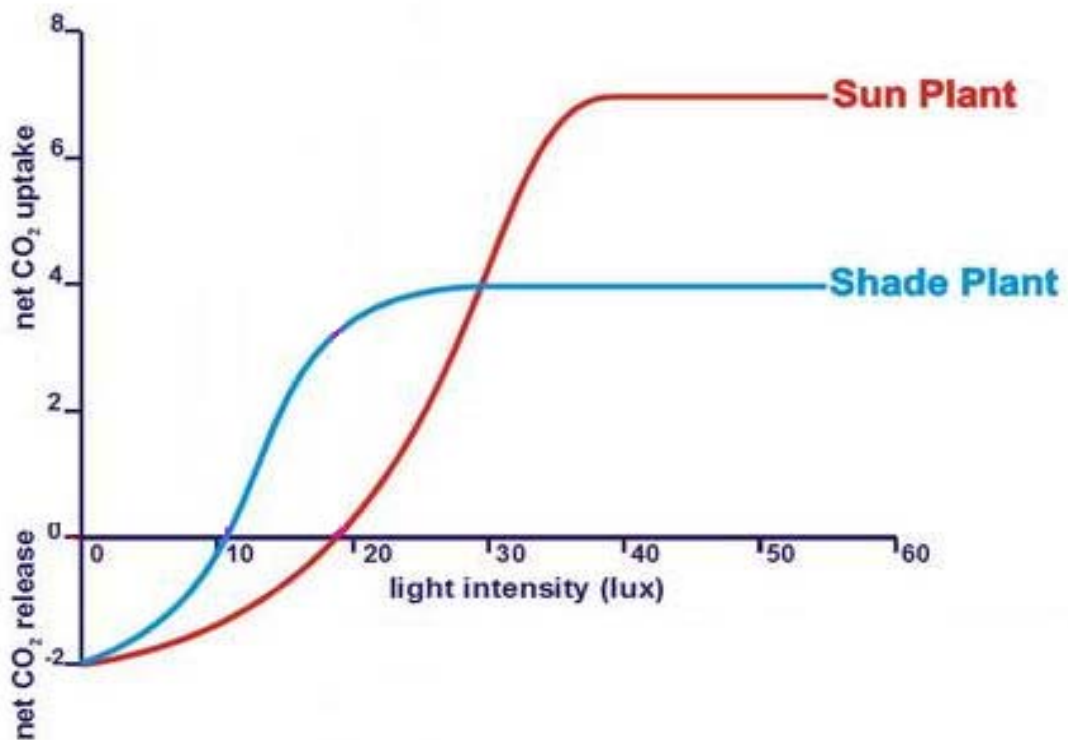


Fig. 3.2

(d) With reference to Fig. 3.2,

(i) state the compensation points of sun and shade plants.

.....
.....[1]

(ii) account for the graph differences between sun and shade plants.

.....
.....
.....
.....
.....
.....
.....[4]

[Total: 15]

4. A study was conducted to study the inheritance of coat colour in mice, in which one of the allele is also known to affect normal embryonic development. A cross between agouti mouse (with agouti coat colour) and yellow mouse (with yellow coat colour) resulted in half of the F1 progeny being agouti mice and the other half being yellow. Mating of F1 yellow mice resulted in the following F2 generation.

Agouti mice 98

Yellow mice 202

- (a) Using the symbols **A** and **a** for the two alleles involved, draw a genetic diagram in the space below to show how the F1 cross resulted in the F2 progeny.

In another experiment involving deer mouse, pure breeding pink-eyed mice with wild-type fur was crossed with pure breeding dark-eyed albino mice. The resulting progeny all had wild-type fur and dark eyes. These F1 mice were then crossed with pink-eyed albino mice. The results are shown in Table 4.1. It was difficult to distinguish between mice that are dark-eyed albino and pink-eyed albino, so these two phenotypes were counted together.

Table 4.1

Phenotype	Number of progeny
Wild-type fur, dark-eyed	12
Wild-type fur, pink-eyed	62
Albino, dark-eyed Albino, pink-eyed	78
Total	152

(b) In the blank space below, calculate the chi-square value.

[2]

Table 4.2 shows a portion of the chi-square table.

Table 4.2

distribution of X^2			
number of degrees of freedom (v)	probability		
	0.1	0.05	0.01
1	2.71	3.84	6.64
2	4.60	5.99	9.21
3	6.25	7.82	11.34
4	7.78	9.49	13.28

- (c) Using the values in Table 4.2, draw appropriate conclusions as to whether the results of the cross followed the expected ratio you have predicted in (b).

.....
.....
.....
.....[2]

- (d) Using **E/e** to represent alleles for eye colour and **A/a** to represent alleles for fur coat colour, explain the result of the F1 cross in the deer mouse experiment using a genetic diagram.

[4]

[Total: 11]

5. *Bungarus multicinctus*, also known as the Taiwanese Krait, is a species of venomous snake endemic to Asia and is predominantly found in forests from Taiwan to Southeast Asia. In order to better understand the venomous snake's physiological pathways, researchers have been conducting extensive research on the mechanism of action of Taiwanese Krait venom which consists primarily of neurotoxins.

Fig. 5.1 shows a diagram of the kappa-bungarotoxin, a neurotoxin found in the venom of the Taiwanese Krait. Kappa-bungarotoxin is a highly stable protein molecule that is capable of withstanding harsh chemical reactions.

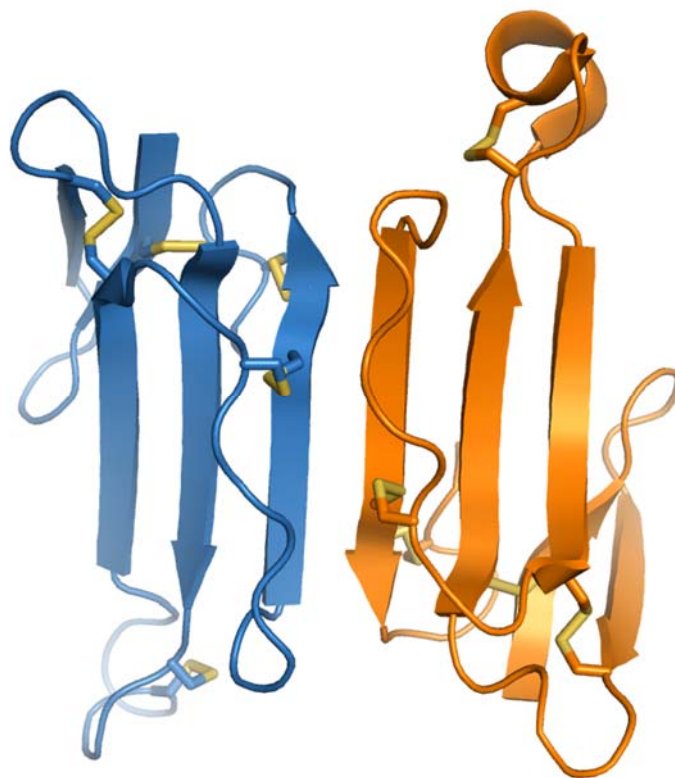


Fig. 5.1

- (a) Explain how the protein structure of kappa-bungarotoxin is maintained.

.....

.....

.....

.....

.....[3]

People bitten by *Bungarus multicinctus* suffer from neuromuscular paralysis and respiratory failure. Research shows that the venom causes serious health complications due to the effects of kappa-bungarotoxin at the neuromuscular junctions at the muscle cells of the lungs as shown in Fig. 5.2.

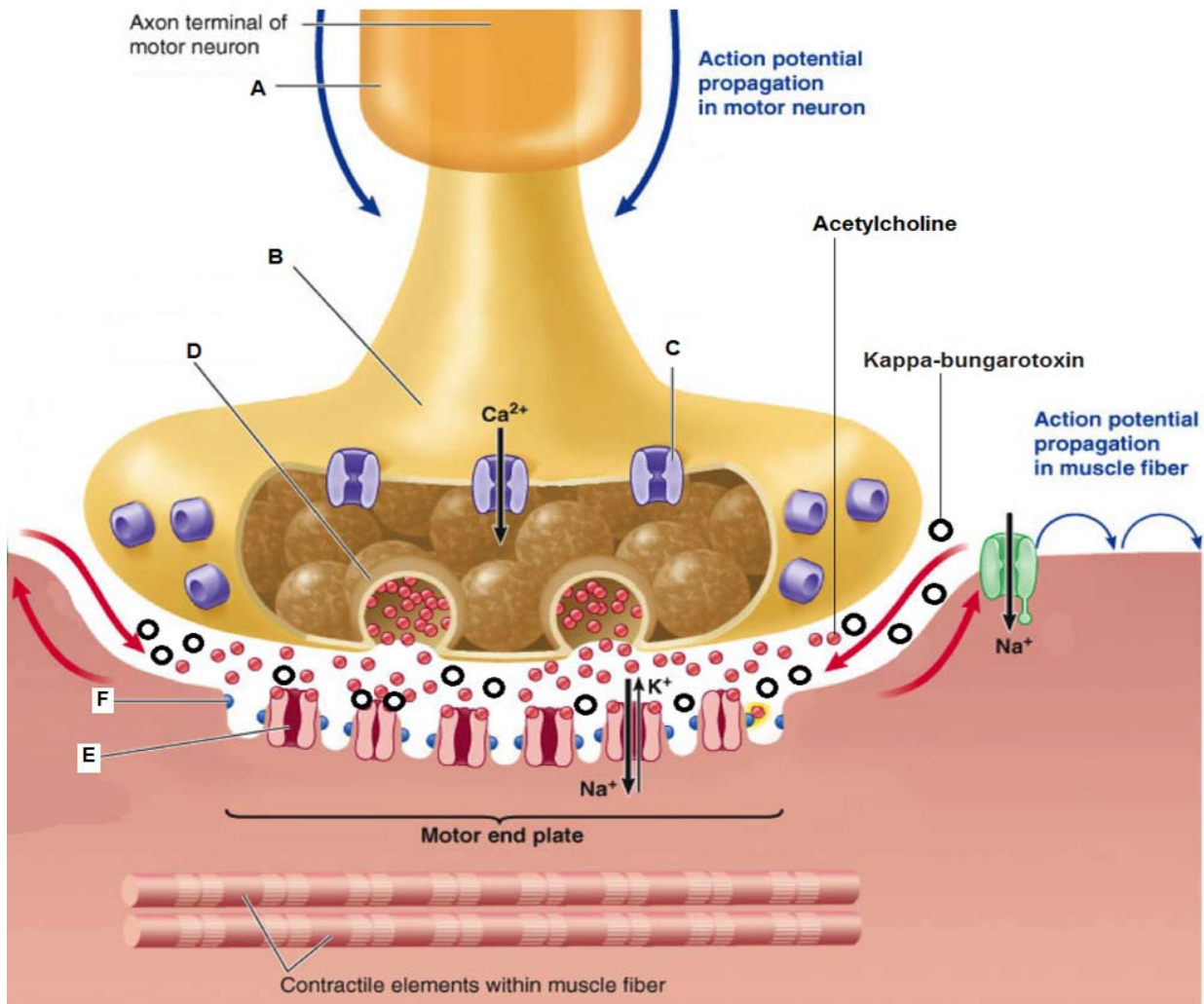


Fig. 5.2

- (b) Name the parts of the neuromuscular junction shown in Fig 5.2 labelled **A**, **B**, **C**, **D** and **E**.

A:

B:

C:

D:

E:

[2]

In Fig. 5.2, **F** is an enzyme important in synaptic signalling.

- (c) Describe the function of **F**.

.....
[1]

- (d) Explain how the transmission of nervous impulse shown in Fig. 5.2 differs from electrical transmission of action potentials.

.....

[2]

- (e) Explain how kappa-bungarotoxin causes respiratory failure among humans bitten by *Bungarus multicinctus*.

.....

[4]

Antivenom is a medication used to treat venomous bites and is recommended for use via injection if the venom from the snake is of high risk of toxicity.

- (f) Suggest how antivenom can alleviate the effects of kappa-bungarotoxin.

.....
[1]

[Total: 13]

6. Fig. 6.1 shows a tyrosine kinase receptor. The effect of insulin binding to this complementary receptor is shown in Fig. 6.2. The boxed regions in Fig. 6.1 and Fig. 6.2 are cysteine-rich domains.

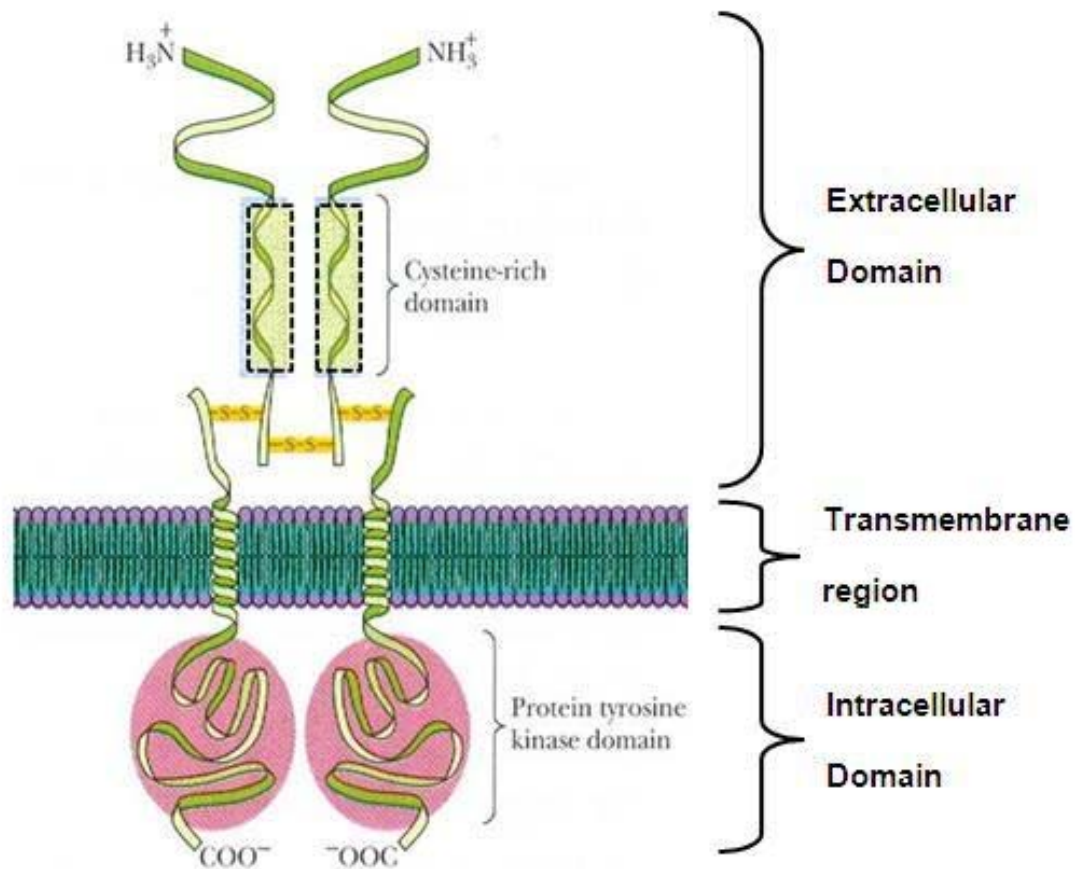


Fig. 6.1

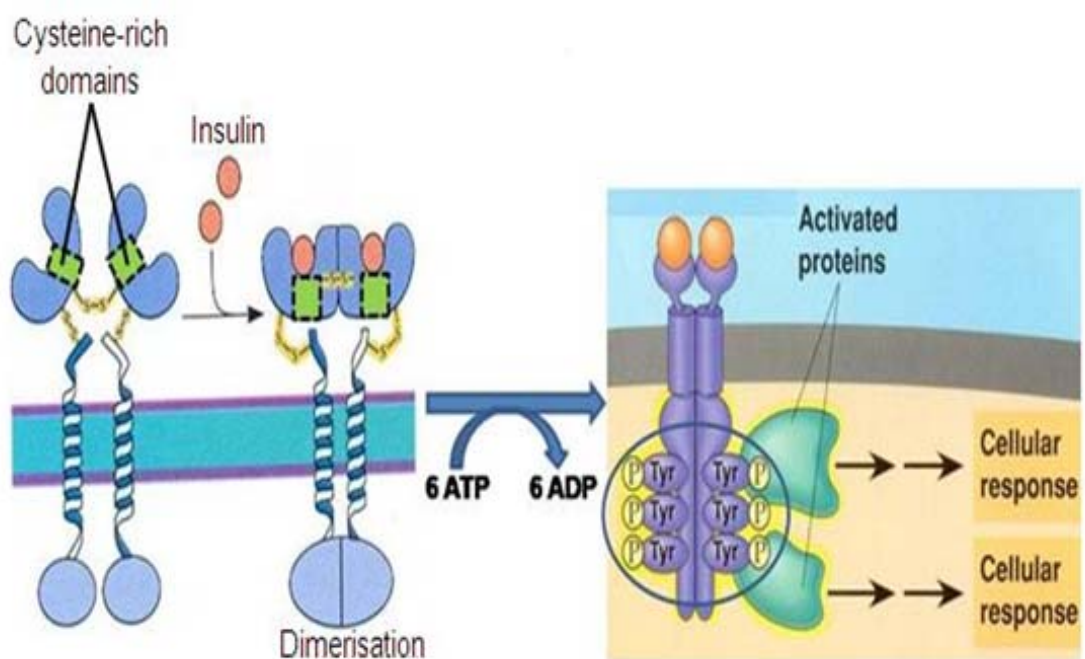


Fig. 6.2

- (a) Explain how the structure of the tyrosine kinase receptor is suited for its role in insulin mediated cell signalling.

.....

[2]

Fig. 6.3 shows how blood levels of glucose, insulin and glucagon change after a meal.

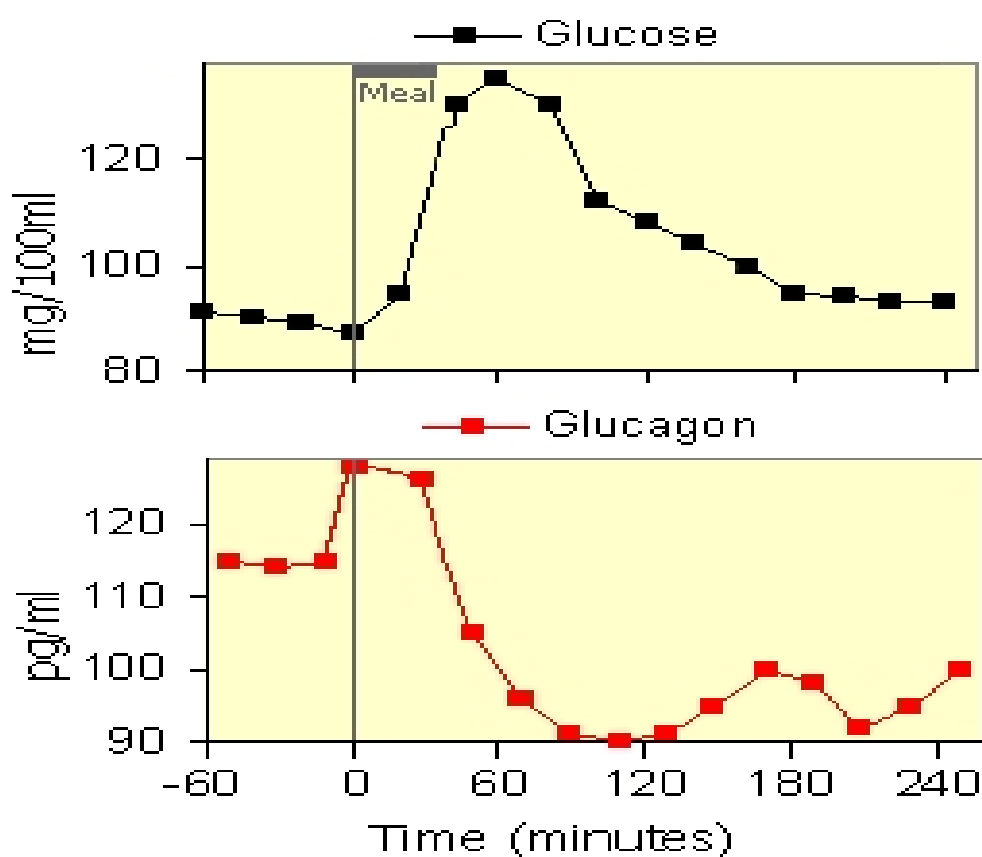


Fig. 6.3

- (b) Describe the components of a homeostatic control system and explain the principles of homeostasis.

.....

.....

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.....

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.....[3]

- (c) With reference to Fig. 6.3, explain the relationship between glucose and glucagon levels from 0 to 120 minutes.

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.....[4]

Diabetes mellitus is a disease in which high blood glucose cannot be regulated back to normal set point within the body. The hormone insulin is commonly used in the treatment of diabetes. There are two forms of diabetes mellitus: Type 1 and Type 2. Type 2 diabetes mellitus is characterized by insulin resistance whereby the body tissues do not respond effectively to insulin.

- (d) Suggest why the body tissue is insensitive to insulin in Type 2 diabetes mellitus.

.....

.....[1]

Glucagonoma is a rare tumour of the α -cells of the islet of Langerhans which results in an overproduction and secretion of glucagon.

- (e) Suggest how glucose metabolism is affected when an individual has a glucagonoma.

.....
.....[1]

Cancer development occurs in stages. The advanced stage of cancer is characterized by the spread of cancer development to other parts of the body via the circulatory system to form secondary tumours by a process known as metastasis.

- (f) Explain the properties of cancer cells required for metastasis.

.....
.....
.....
.....
.....[2]

[Total: 13]

7. In New Zealand, there are two species and three sub-species of native bush robins. It is believed that the robins evolved from a common ancestral stock, members of which flew from Australia across the Tasman Sea and became established in New Zealand over a million years ago. This ancestral form is considered to be similar to the present day Australian flame robin, *Petroica multicolor*, a bird with a brightly coloured red breast. The New Zealand birds do not have this red colour. Some characteristics of the birds and their distributions are shown in Fig. 7.1.

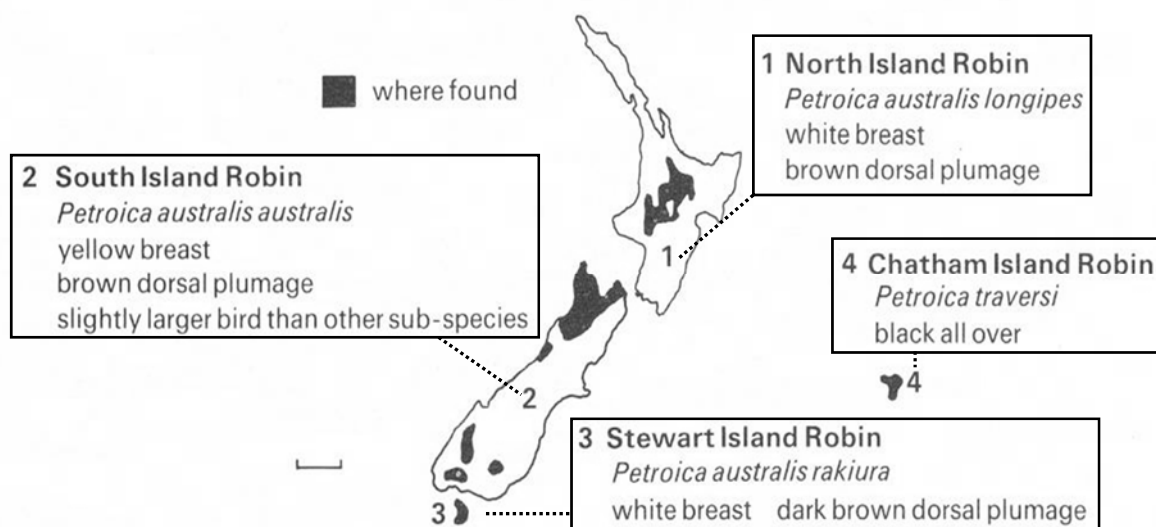


Fig. 7.1

- (a) State the scientific name of the two species of native bush robins in New Zealand.

..... [1]

- (b) Explain why the robins in locations 1, 2 and 3 are similar but different from those in location 4.

.....

.....

.....

.....

.....

.....[3]

Based on earlier research, the native bush robin species and subspecies were distinguished based on a number of phenotypic differences such as plumage and breast colour. However, with modern technology, researchers have been using molecular methods such as direct DNA and amino acid comparison to further determine the phylogeny between the native bush robin species.

- (c) Explain why molecular homology is better than anatomical homology in determining evolutionary relationship between species of native bush robins.

.....

[2]

Differences in the *cytochrome b* DNA sequence of several native bush robins from different regions of New Zealand were measured and plotted against time since divergence from the primitive ancestor as seen in Fig. 7.2.

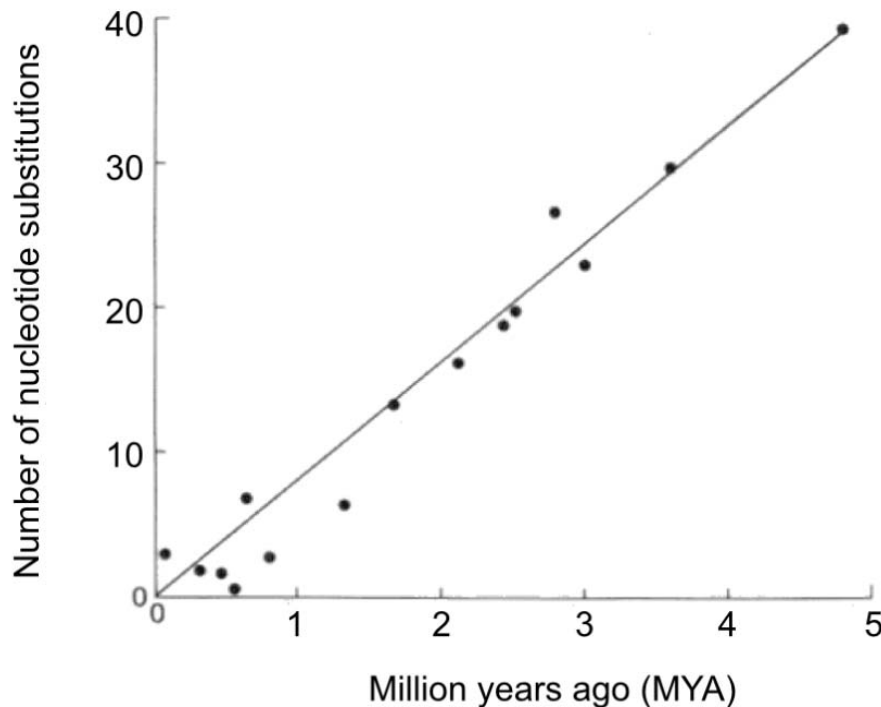


Fig. 7.2

- (d) Describe how the differences in the number of nucleotide substitutions support the neutral theory of molecular evolution.

.....
.....
.....
.....[2]

- (e) Suggest why New Zealand robins do not have the red breast trait even though it is present in its Australian robin ancestors.

.....
.....[1]

[Total: 9]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

8.

- (a)** Compare between DNA replication and transcription. **[6]**
- (b)** Describe how differences in the structure and organization of prokaryotic and eukaryotic genomes affect their control of gene expression. **[7]**
- (c)** Outline the viral reproduction cycle of HIV. **[7]**

[Total: 20]

9.

- (a)** Contrast the structures of viral, prokaryotic and eukaryotic genome. **[5]**
- (b)** Relate the structure of ribosome to its role in protein synthesis. **[7]**
- (c)** Outline the processes involved in oxidative phosphorylation. **[8]**

[Total: 20]

End of Paper